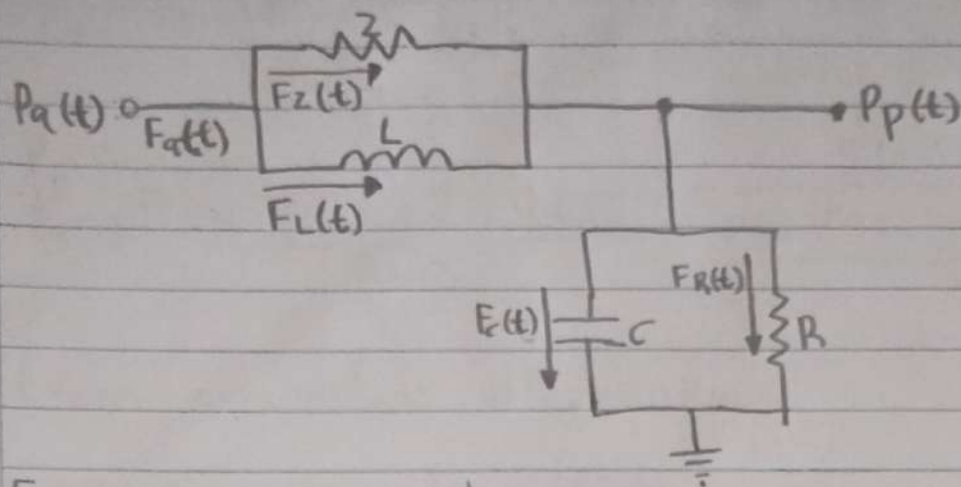


Practica 5.4: Sistema cardiovascular



Ecuación principal:

$$F_a(t) = F_z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_C(t) = C \frac{dP_p(t)}{dt}$$

$$F_L(t) = \frac{1}{L} \int [P_a(t) - P_p(t)] dt$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Procedimiento algebraico:

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt = C \frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{Ls}\right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right) P_p(s)$$

$$\frac{Ls+Z}{LZs} P_a(s) = \frac{CLRZs^2 + LZs + RLs + RZ}{RLZs} P_p(s)$$

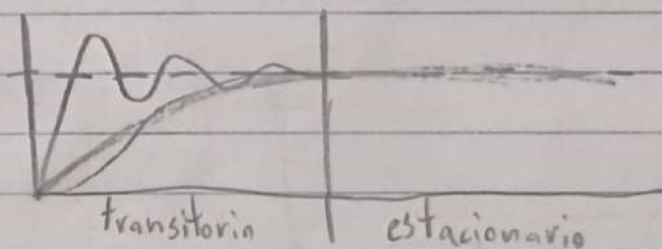
$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + (LZ + RL)s + RZ}{RLZs}} = \frac{R(Ls+Z)}{CLRZs^2 + (LZ + RL)s + RZ}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s P_a(s) \left[1 - \frac{P_p(s)}{P_a(s)} \right]$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{RLs + RZ}{CLRZs^2 + (LZ + RL)s + RZ} \right]$$

$$= 1 - \frac{RZ}{RZ} = 0V$$



Estabilidad en Lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{aligned} a &= CLRZ & \lambda_{1,2} &= \frac{-(LZ + RL) \pm \sqrt{(LZ + RL)^2 - 4CLRZ^2}}{2CLRZ} \\ b &= LZ + RL \\ c &= RZ \end{aligned}$$

Respuesta Estable porque $\text{Re} \lambda_{1,2} < 0$

Modelo de ecuaciones integradiferenciales

$$P_p(t) \left(\frac{1}{R} + \frac{1}{Z} \right) = \frac{P_a(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt - \frac{C d P_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt - \frac{C d P_p(t)}{dt} \right) \frac{ZR}{Z+R}$$

